TrustEstate

Requirements Specification

3) Requirements

a) Functional Requirements

The following table defines the functional requirements of the TrustEstate platform, derived from the user scenarios and use cases defined in Section 4. Each requirement is assigned a unique identifier, a priority level (1 = Highest, 3 = Lowest), and the date reviewed.

Priority Scale: 1 = Must Have (Highest) | 2 = Should Have | 3 = Nice to Have (Lowest)

Req#	Requirement	Comments	Priority	Date Rvwd
FR_01	The system shall provide role-specific dashboards for each user type: Admin, Agent, Property Inspector, Property Owner, and Buyer/Tenant.	Every role has its own dashboard with features and views relevant to their responsibilities.	1	
FR_02	User passwords must be stored securely in the database.	Passwords will be saved as encrypted hashes rather than plain text to ensure data security.	1	
FR_03	The system shall authenticate users using registered email and password credentials.	Upon login, the system validates credentials and redirects the user to their role-specific dashboard.	1	
FR_04	The system shall lock or flag an account after a defined number of consecutive failed login attempts.	Failed login attempts must be logged, and repeated failures must trigger protective account measures.	1	
FR_05	Admin shall be able to review pending Agent and Property Inspector accounts and approve or reject them within 72 hours.	Admin reviews professional details and agency affiliation; approval grants a Verified badge, rejection notifies the user with a documented reason.	1	
FR_06	Admin shall be able to monitor all active, flagged, and suspended listings on the platform.	Admin may flag, suspend, or remove any listing that violates platform policies, with notifications sent to the Agent and Property Owner.	1	
FR_07	Admin shall be able to monitor all submitted inspection reports for quality and compliance.	Admin may flag or remove fraudulent or non-compliant reports; all actions are logged with a documented reason.	2	
FR_08	Admin shall be able to manage all user accounts, including deactivating, suspending, or reactivating them.	Status changes take effect immediately and prevent suspended/deactivated users from logging in; affected users are notified.	1	
FR_09	Admin shall be able to manage disputes by reviewing listing details,	Admin may mark a dispute as resolved with a documented outcome	2	

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	offer history, inspection reports, and communications between parties.	or temporarily suspend a transaction if a party is unresponsive.		
FR_10	Admin shall have access to a real-time analytics dashboard displaying platform-wide metrics.	Dashboard shows active listings, flagged listings, pending verifications, ongoing transactions, registered users per role, and open disputes.	2	
FR_11	Property Inspectors and Agents shall self-register through the public registration page; their accounts shall be created in Pending Verification status and require Admin approval before activation.	Property Inspectors provide personal and professional details; Agents provide agency affiliation (Independent or Agency Name). Admin must review and approve or reject within 72 hours.	1	
FR_12	Upon Admin approval, the system shall assign a Verified badge to the approved account and notify the user.	The Verified badge must appear immediately upon account activation. Only Admin-approved accounts receive the badge.	1	
FR_13	Property Owners and Buyers/Tenants shall self-register and their accounts shall be activated immediately without Admin approval.	No professional verification is required for these roles; accounts become active immediately after registration.	1	
FR_14	Verified Property Inspectors shall be able to view all inspections assigned to them, including property details, scheduled date, and the assigned Agent's contact information.	Inspectors can only view inspections explicitly assigned to them.	1	
FR_15	Property Inspectors shall be able to update the inspection status in sequence: Scheduled → In Progress → Completed.	Status updates may only be performed on the scheduled day; the system must enforce the correct sequence and prevent out-of-order updates.	1	
FR_16	Property Inspectors shall submit a structured inspection report upon completing an inspection, covering four fixed categories: Structural Integrity, Plumbing, Electrical, and Safety.	Each category requires written findings, a pass/fail rating, a severity classification (Minor, Moderate, or Critical), and supporting photos.	1	
FR_17	Once formally submitted, an inspection report shall be permanently locked and cannot be modified or resubmitted under any circumstance.	Report locking applies immediately upon submission and is enforced by the system with no manual override.	1	
FR_18	Property Inspectors shall assign a final verdict of Passed, Passed with Conditions, or Failed to each completed inspection.	The verdict triggers automatic notifications to the Buyer/Tenant, Agent, and Property Owner, and grants them access to the full report.	1	
FR_19	Agents shall be able to accept or decline listing assignment requests from Property Owners.	Upon acceptance, the listing status updates to Pending Agent Review. Upon decline, the Property Owner is notified to select a different Agent.	1	
FR_20	Agents shall be able to review submitted listings in full detail and	Approved listings are set to Active status and become publicly visible;	1	

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	either approve them or request corrections from the Property Owner.	correction requests change the status to Corrections Requested.		
FR_21	Agents shall manage the full offer lifecycle on behalf of the Property Owner, including accepting, declining, or countering offers.	A maximum of 3 negotiation rounds is enforced per offer. Counter-offers include a response deadline enforced by the system.	1	
FR_22	Agents shall assign an available verified Property Inspector to a listing once an offer is accepted, and may reassign a replacement Inspector if needed.	Only verified Property Inspectors appear in the selection list. The system notifies all relevant parties upon assignment.	1	
FR_23	Agents shall be able to close a transaction once all conditions are met: accepted offer, completed inspection, and no pending disputes.	Upon closure, the listing is archived and all relevant parties are notified. The system prevents closure if any condition is unmet.	1	
FR_24	Agents shall be able to message Buyers/Tenants who have submitted offers on their assigned listings through the platform's internal messaging system.	Messaging is restricted to Buyers/Tenants with active or historical offers on the Agent's assigned listings.	2	
FR_25	Property Owners shall be able to create and submit a property listing with a title, description, location, asking price, listing type (sale/rent), and photos.	Owners must select a verified Agent to manage the listing. The listing is created in Pending Agent Review status.	1	
FR_26	Property Owners shall be able to edit listing details while the listing is in Pending Agent Review or Corrections Requested status.	Active listings cannot be edited directly by the owner; the owner must contact the assigned Agent to request changes.	1	
FR_27	Property Owners shall be able to remove a listing at any time provided no active transaction is in progress.	The system checks for active transactions in real time before permitting removal.	1	
FR_28	Property Owners shall be able to view all offers submitted on their listings, including proposed prices, messages, and current statuses.	Owners receive notifications for new offers and status changes. Offer actions (accept/decline/counter) are handled exclusively by the assigned Agent.	2	
FR_29	Property Owners shall be able to monitor the statuses of all their listings and receive automatic notifications at every relevant stage of a transaction.	The dashboard displays current statuses for all owner listings; notifications are delivered at each key transaction milestone.	2	
FR_30	Property Owners shall be able to view the full completed inspection report for any of their properties.	Owners receive a notification when the report is submitted and can review all category findings, severity levels, photos, and the final verdict.	2	
FR_31	Property Owners shall be able to communicate with the assigned Agent regarding their listing or	Messaging is restricted to the Agent assigned to the owner's listing.	2	

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	ongoing transaction through the platform's messaging system.			
FR_32	Buyers/Tenants shall be able to browse all active listings and filter them by location, price range, property type, and listing type (sale/rent) without requiring login.	Listing browsing and filtering is publicly accessible; login is required only for offer submission and saving favourites.	1	
FR_33	Buyers/Tenants shall be able to view a detailed listing page including photos, description, location, price, listing type, current status, and the assigned Agent's profile.	Listing details are publicly accessible; login is required to save or submit an offer.	1	
FR_34	Buyers/Tenants shall be able to save listings to a personal favourites list.	Users can add or remove listings from their favourites at any time after login.	3	
FR_35	Buyers/Tenants shall be able to submit an offer on an active listing with a proposed price and an optional message.	Only one active offer per listing per user is permitted. The assigned Agent and Property Owner are notified upon submission.	1	
FR_36	Buyers/Tenants shall be able to participate in offer negotiation, responding to counter-offers by accepting, declining, or submitting a revised offer.	Negotiation is limited to 3 rounds per offer; unresolved offers expire automatically and the listing returns to Active status.	1	
FR_37	Buyers/Tenants shall be able to withdraw an active offer at any time before it is accepted.	Upon withdrawal, the offer status is updated and the assigned Agent is notified.	2	
FR_38	Buyers/Tenants with an accepted offer shall be able to view the full inspection report once submitted, including findings, severity classifications, photos, and the final verdict.	Access to the full inspection report is granted automatically upon the Inspector assigning the final verdict.	1	
FR_39	Following a Failed or Passed with Conditions inspection verdict, Buyers/Tenants shall be able to withdraw their offer or submit a revised offer within a defined response window.	System notifies the user of available options within the response window and processes the selected action accordingly.	1	
FR_40	Buyers/Tenants shall be able to message the Agent assigned to a listing on which they have submitted an offer through the platform's internal messaging system.	Messaging is restricted to Agents managing listings on which the Buyer/Tenant has submitted an offer.	2	
FR_41	The system shall enforce real-time form validation across all registration and submission forms, highlighting errors before submission.	Validation must execute client-side in real time without requiring a full page reload.	1	
FR_42	All notifications triggered by system events shall be delivered to relevant	Notifications apply across all event types: account decisions, listing status	1	

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	parties within 1 minute of the triggering event.	changes, offer responses, inspection updates, and messages.		
FR_43	The system shall log all key actions with timestamps for audit purposes, including login attempts, verification decisions, listing actions, offer events, inspection updates, and dispute resolutions.	Logs must be retained and accessible for administrative review and dispute resolution.	1	
FR_44	The web application shall be responsive and accessible from any device.	The interface must adapt to desktop, tablet, and mobile screen sizes without loss of functionality.	1	
FR_45	The system shall implement secure authentication and authorization, ensuring each user can only access features and data relevant to their role.	Role-based access control must be enforced at the system level; users cannot access or modify data outside their permitted scope.	1	